

Critical Euler Errors for Harmonic Polylogarithms

Sharp constants, positive surplus kernels, and a drifting extremizer

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Abstract

Let $F_{a,b}(z) = \sum_{n \geq 1} H_{n-1}^{(b)} z^n / n^a$, with principal continuation to $\mathbb{C} \setminus [1, \infty)$, and let $g_{a,b} = \operatorname{Im} F_{a,b}(i)$. We prove the sharp-critical-constant conjecture in the incoming report *The Sharp Real-Order Threshold for Harmonic Polylogarithms*: $a \mapsto -2g_{a,1-a}$ is strictly decreasing on $(0, 1)$, and the optimal uniform constant in its one-sided Euler bound is $C_* = \pi/4 + (\log 2)/2$. The proof comes from a fixed-total-order beta-gamma transport. It gives strict order monotonicity for every total order $a+b > 0$ at imaginary radii up to $\sqrt{3}$, as well as exact endpoint comparisons on the real slit interval.

A second argument proves that the critical negative density is strictly larger than Lebesgue density. The resulting positive surplus has an exact balanced Bose-kernel representation, strengthens the harmonic-defect Hausdorff moment inequalities, and yields a zero-free Stieltjes factorization for a difference of polylogarithms.

For the scaled Euler error $R_N(a) = 2^N(E_N - g_{a,1-a})$, we establish a uniform asymptotic and determine its moving maximizer. With $L = \frac{1}{2} \log N$ and $\ell = \log L$, every maximizing outer order a_N satisfies $(1 - a_N)\ell \rightarrow 1$, while $\max_a R_N(a) = \frac{\sqrt{\pi}}{2\sqrt{N}}[1 + (1 + o(1))/(eL\ell)]$. A refinement identifies a coefficient involving the first Stieltjes constant. Exact rational certificates demonstrate the finite-depth reversal $R_{128}(1/2) > R_{128}(1/4)$. Thus the proved monotonicity of the sharp all-depth constant does not extend to each individual scaled error. All analytic assertions are proved in the text; exact arithmetic, numerical diagnostics, and remaining conjectures are explicitly separated.

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1 Scope, baseline, and the question resolved

The canonical ProveIt manuscript *Polylogarithms and their Arithmetic Bridges* develops signed kernels and certified evaluation alongside cyclotomic reductions and arithmetic special values [1]. The incoming report *The Sharp Real-Order Threshold for Harmonic Polylogarithms* extends the finite signed moment representation to the sharp domain $a + b \geq 1$ and identifies the required atom on its boundary [2]. Its Conjecture 12.3, labelled `conj:constant` in the delivered source, asks for the strict decrease of $-2g_{a,1-a}$ and the optimal uniform Euler constant. We settle that conjecture and then examine an optimization problem it does not settle: which parameter is worst at a prescribed finite Euler depth?

The repository revision inspected for this delivery is

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The finite-measure threshold, the basic double kernel, critical angular uniqueness, and the fixed-parameter Euler remainder formula below are *prior results*. We include short derivations of the identities needed here so that the new arguments can be checked independently. The new results are the fixed-total-order comparisons, proof of the named conjecture, strict surplus density, strengthened moment consequences, uniform critical error expansion, and extremizer drift. We do not claim global literature priority for every identity, and none of the analytic proofs has been formalized in a proof assistant or independently refereed. The classical beta, gamma, and zeta facts are cited at their point of use.

1.1 Notation and summation conventions

For real $a, b > 0$, define

$$c_n(a, b) = \frac{H_{n-1}^{(b)}}{n^a}, \quad H_m^{(b)} = \sum_{j=1}^m j^{-b}, \quad F_{a,b}(z) = \sum_{n=1}^{\infty} c_n(a, b) z^n \quad (|z| < 1). \quad (1.1)$$

Thus $c_1 = 0$ and $c_2 = 2^{-a}$. At positive integer orders, $F_{a,b}(z) = \text{Li}_{a,b}(z, 1)$ in the decreasing-index convention. We write

$$G_{a,b}(\rho) = -\text{Im } F_{a,b}(i\rho), \quad g_{a,b} = \text{Im } F_{a,b}(i), \quad w = a + b.$$

The principal polylogarithm conventions are those of DLMF 25.12 [5]. On $a + b = 1$ the coefficients tend to $1/a$, so the ordinary boundary series fails the term test. All Gaussian values on that line mean the analytic continuation, equivalently the Abel value. Euler transformation below is justified by a measure identity, not by pretending the original boundary series converges.

Unless stated otherwise, a constant implicit in $O(\cdot)$ may depend on fixed parameters. The word *uniform* always identifies the parameter set over which the constant and the threshold are independent. We use $\Delta Q_n = Q_n - Q_{n+1}$ for the decreasing finite-difference convention.

2 Fixed-total-order beta–gamma transport

2.1 A positive double kernel and a different coupling

The useful part of the earlier double integral is not merely its positivity. A change of variables makes the total order, rather than the two separate orders, the gamma shape.

Lemma 2.1 (Double kernel, recalled with proof). *For $a, b > 0$ and $z \in \mathbb{C} \setminus [1, \infty)$,*

$$F_{a,b}(z) = z^2 \int_{0 < y < x < 1} \frac{(-\log x)^{a-1} (\log(x/y))^{b-1}}{\Gamma(a)\Gamma(b)(1-zx)(1-zy)} dy dx. \quad (2.1)$$

The positive weight, without the two denominator factors, has mass 2^{-a} .

Proof. Expand the denominators for $|z| < 1$. For nonnegative integers j, k , substitution $y = xv$ and two gamma integrals give

$$\int_{0 < y < x < 1} \frac{(-\log x)^{a-1} (\log(x/y))^{b-1}}{\Gamma(a)\Gamma(b)} x^j y^k dy dx = \frac{1}{(j+k+2)^a (k+1)^b}.$$

Summation over $j+k = n-2$ yields c_n . Absolute convergence justifies the expansion, and $j=k=0$ gives the mass. On compact subsets of the slit plane both denominators stay uniformly away from zero. Dominated holomorphic integration then continues the identity to the claimed domain. $\square$

Theorem 2.2 (Fixed-weight transport identity). *Let $T \sim \text{Gamma}(w, 1)$ and $V \sim \text{Beta}(a, b)$ be independent, where $w = a + b$. Put $X = e^{-TV}$ and $Y = e^{-T}$. For $z \in \mathbb{C} \setminus [1, \infty)$,*

$$\frac{F_{a,b}(z)}{z^2} = \mathbb{E} \left[\frac{X}{(1-zX)(1-zY)} \right], \quad (2.2)$$

with the quotient continuously extended at zero. In particular,

$$G_{a,b}(\rho) = \rho^3 \mathbb{E}[H_\rho(X, Y)], \quad H_\rho(x, y) = \frac{x(x+y)}{(1+\rho^2 x^2)(1+\rho^2 y^2)}. \quad (2.3)$$

Proof. In (2.1), set $U = -\log x$, $S = \log(x/y)$, $T = U + S$, and $V = U/T$. Then $x = e^{-TV}$, $y = e^{-T}$, and

$$dx dy = e^{-2U-S} dU dS, \quad dU dS = T dT dV.$$

Since $e^{-2U-S} = e^{-T} e^{-TV}$, the resulting measure is X times

$$\frac{e^{-T} T^{w-1}}{\Gamma(w)} dT \frac{V^{a-1} (1-V)^{b-1}}{B(a, b)} dV.$$

Euler's beta identity $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ [3] proves (2.2). The imaginary part at $z = i\rho$ is obtained by rationalizing the two denominator factors. $\square$

Remark 2.3 (The extra factor is essential). The expectation in (2.2) has the numerator X . Omitting it would produce the wrong coefficients and invalidate the monotonicity argument. The earlier normalization using independent gamma variables at rates two and one instead has a prefactor 2^{-a} . The coupling here uses rate one for the common total T and keeps the extra factor inside the expectation. This is a change of representation, not a correction to the earlier correctly normalized formula.

2.2 Strict stochastic comparison

Lemma 2.4 (Beta score covariance). *Fix $w > 0$. If $V_a \sim \text{Beta}(a, w - a)$, $0 < a < w$, and f is bounded and strictly decreasing on $(0, 1)$, then $\mathbb{E}f(V_a)$ is strictly decreasing in a . On compact subintervals of $(0, w)$,*

$$\frac{d}{da} \mathbb{E}f(V_a) = \text{Cov} \left(f(V_a), \log \frac{V_a}{1 - V_a} \right) < 0. \quad (2.4)$$

The derivative formula also holds for bounded measurable f without a strictness assertion.

Proof. Differentiate the normalized beta density. Its score is the centered version of $\log(v/(1 - v))$, so differentiation gives (2.4). The logarithmic factors are integrable uniformly on compact shape-parameter intervals. If V'_a is an independent copy, then the covariance equals

$$\frac{1}{2} \mathbb{E} \left[(f(V_a) - f(V'_a)) \left(\log \frac{V_a}{1 - V_a} - \log \frac{V'_a}{1 - V'_a} \right) \right].$$

The product is strictly negative whenever the two variables differ. Their densities are positive and continuous in the interior, so equality has probability zero. This proves strictness. $\square$

Theorem 2.5 (Fixed-total-order comparisons). *For every $w > 0$ and $0 < \rho \leq \sqrt{3}$, the function*

$$a \mapsto G_{a, w-a}(\rho), \quad 0 < a < w, \quad (2.5)$$

is strictly decreasing. For every real $z < 1$, the function $a \mapsto F_{a, w-a}(z)/z^2$ is also strictly decreasing, with its continuous interpretation at $z = 0$.

Proof. For $0 < y < x < 1$, direct differentiation gives

$$\partial_x H_\rho(x, y) = \frac{2x + y - \rho^2 x^2 y}{(1 + \rho^2 y^2)(1 + \rho^2 x^2)^2} > 0. \quad (2.6)$$

Indeed, for $\rho^2 \leq 3$ the numerator is at least $2x + y - 3x^2 y \geq 3y(1 - x^2) > 0$. For fixed $T > 0$, $Y = e^{-T}$ is fixed and $X = e^{-TV}$ decreases strictly with V . Thus $H_\rho(e^{-TV}, e^{-T})$ is bounded and strictly decreasing in V . Apply Lemma 2.4 conditionally on T and then integrate its strict inequality against the common gamma law. This proves (2.5).

For real $z < 1$, replace H_ρ by $x/((1 - zx)(1 - zy))$. Its x -derivative is

$$\frac{1}{(1 - zy)(1 - zx)^2} > 0.$$

The same proof applies. At $z = 0$, the quotient is 2^{-a} , consistently with the continuous extension. All integrands used here are uniformly bounded for the fixed values of z or ρ in the theorem. $\square$

The radius $\sqrt{3}$ is a sufficient domain obtained from the pointwise kernel inequality. We do not assert that it is the maximal domain of integrated order monotonicity. When $\rho > \sqrt{3}$, the numerator in (2.6) can be negative near $x = y = 1$, but that fact alone is not a counterexample for the expectation.

2.3 Endpoint identities and explicit Gaussian bounds

Proposition 2.6 (Order-zero endpoints). *For fixed $w > 0$, locally uniformly for $z \in \mathbb{C} \setminus [1, \infty)$,*

$$\lim_{a \downarrow 0} F_{a,w-a}(z) = \frac{z}{1-z} \text{Li}_w(z), \quad (2.7)$$

$$\lim_{a \uparrow w} F_{a,w-a}(z) = \text{Li}_{w-1}(z) - \text{Li}_w(z). \quad (2.8)$$

Proof. The beta variable tends in probability to zero as $a \downarrow 0$ and to one as $a \uparrow w$: its respective means or complementary means tend to zero. Bounded convergence in (2.2) therefore supplies both limiting holomorphic functions, uniformly on compact slit sets. In the disk their coefficients are respectively $H_{n-1}^{(w)}$ and $(n-1)n^{-w}$. Summing the first sequence by a geometric series and splitting the second proves the displayed formulas. $\square$

Define $\beta(s) = \text{Im Li}_s(i)$ and $\eta(s) = -\text{Li}_s(-1)$ by continuation for real s . For positive w the second quantity is the usual Dirichlet eta value. The parity split in the absolutely convergent disk series, continued to $z = i$, gives $\text{Re Li}_w(i) = -2^{-w}\eta(w)$. Consequently Theorem 2.5 implies the strict bounds

$$\beta(w) - \beta(w-1) < -g_{a,b} < \frac{\beta(w) + 2^{-w}\eta(w)}{2}, \quad a, b > 0, \quad a + b = w. \quad (2.9)$$

Both endpoints are sharp as limiting parameter values. On the real interval $z < 1$, the analogous strict comparison is

$$\frac{\text{Li}_{w-1}(z) - \text{Li}_w(z)}{z^2} < \frac{F_{a,b}(z)}{z^2} < \frac{\text{Li}_w(z)}{z(1-z)}, \quad (2.10)$$

with continuous values at zero. These are exact analytic inequalities, not numerical integer-relation discoveries.

3 The critical measure and the sharp Euler constant

3.1 The critical representation

From now on $0 < a < 1$, $b = 1 - a$. Define

$$k_a(T) = \frac{T^{a-1}}{\Gamma(a)}, \quad h(t) = \frac{1}{1-e^{-t}} - \frac{1}{t}. \quad (3.1)$$

The function h increases from $1/2$ to 1 , since $h'(t) = t^{-2} - [4 \sinh^2(t/2)]^{-1} > 0$. Put

$$C_a(T) = \frac{1}{\Gamma(a)\Gamma(b)} \int_0^T (T-t)^{a-1} t^{b-1} h(t) dt, \quad (3.2)$$

$$p_a(T) = -\zeta(b)k_a(T) + C_a(T). \quad (3.3)$$

Here $C_a(T) = \mathbb{E}h(TV)$ for $V \sim \text{Beta}(b, a)$; note that the order of these beta parameters differs from the transport variable in Section 2. In particular $1/2 < C_a(T) < 1$.

Lemma 3.1 (Critical signed measure, prior identity). *The density $p_a(-\log u)$ is positive, has mass $1/a$ on $(0, 1)$, and*

$$\mu_a = \frac{1}{a} \delta_1 - p_a(-\log u) du \quad (3.4)$$

has moments $\int u^{n-1} d\mu_a = c_n(a, 1-a)$. Thus

$$F_{a,1-a}(z) = \int_{[0,1]} \frac{z}{1-zu} d\mu_a(u) \quad (z \in \mathbb{C} \setminus [1, \infty)). \quad (3.5)$$

Proof. The regularized Hurwitz integral [4] is

$$\zeta(b, q) = \frac{q^{1-b}}{b-1} + \frac{1}{\Gamma(b)} \int_0^\infty e^{-qt} t^{b-1} h(t) dt \quad (0 < b < 1, q > 0).$$

It follows by splitting the standard integral for $b > 1$ and continuing the bounded-regularizer integral to $\Re b > 0$. Taking the Laplace transform of (3.2), and using $a + b = 1$, gives

$$\int_0^\infty e^{-qT} p_a(T) dT = \frac{1}{a} - q^{-a}(\zeta(b) - \zeta(b, q)). \quad (3.6)$$

The finite Hurwitz difference at $q = n$ is $H_{n-1}^{(b)}$. The case $n = 1$ gives the mass $1/a$. Positivity follows from $\zeta(b) < 0$, $k_a > 0$, and $C_a > 0$. One may obtain the zeta sign from its positive alternating eta series and $1 - 2^{1-b} < 0$. Expansion of the resolvent in the disk and dominated continuation prove (3.5). This reproduces the critical part of [2, Theorem 3.1], not a new threshold assertion. $\square$

3.2 Euler errors and the conjecture

Define, exactly as in the incoming report,

$$f(n) = c_{2n+1}(a, 1-a), \quad d_k = \sum_{n=0}^k (-1)^n \binom{k}{n} f(n), \quad E_N = \sum_{k=0}^{N-1} \frac{d_k}{2^{k+1}}. \quad (3.7)$$

For $N \geq 1$, let

$$R_N(a) = 2^N(E_N - g_{a,1-a}), \quad q_N(u) = \frac{(1-u^2)^N}{1+u^2}. \quad (3.8)$$

The binomial theorem, followed by a geometric summation under the finite signed measure, gives

$$R_N(a) = \int_0^1 q_N(u) p_a(-\log u) du > 0. \quad (3.9)$$

The endpoint atom contributes zero because $N \geq 1$. For fixed a , these integrals decrease strictly to zero, and

$$R_1(a) = -2g_{a,1-a}, \quad 0 < E_N - g_{a,1-a} < \frac{1}{a} 2^{-N}. \quad (3.10)$$

These facts were already established in [2, Section 9].

Theorem 3.2 (Resolution of the sharp-critical-constant conjecture). *The map $a \mapsto R_1(a) = -2g_{a,1-a}$ decreases strictly on $(0, 1)$. Its endpoint limits are*

$$R_1(0) = \frac{\pi}{4} + \frac{\log 2}{2} = C_*, \quad R_1(1) = \frac{\pi}{2} - 1. \quad (3.11)$$

For every $0 < a < 1$ and every integer $N \geq 1$,

$$0 < E_N - g_{a,1-a} < C_* 2^{-N}. \quad (3.12)$$

No smaller constant works uniformly over this open parameter interval and all such N .

Proof. Set $w = \rho = 1$ in Theorem 2.5. At the two order endpoints use (2.7)–(2.8), $\text{Li}_1(z) = -\log(1-z)$, and $\text{Li}_0(z) = z/(1-z)$. Equivalently, (2.9) uses $\beta(1) = \pi/4$, $\beta(0) = 1/2$, and $\eta(1) = \log 2$. This proves (3.11) and strict decrease. Since $R_N(a) \leq R_1(a) < C_*$, (3.12) follows. If $C < C_*$, choose a sufficiently close to zero that $R_1(a) > C$; then $N = 1$ contradicts the proposed bound. Thus the constant is sharp. $\square$

Numerically $C_* = 1.13197175367742096432427690655\dots$. The theorem proves both components of the prior conjecture, not just an upper estimate. The distinction between a strict uniform bound on an open parameter domain and an attained maximum matters: C_* is a supremum at $a = 0, N = 1$, not a maximum with $0 < a < 1$. For one fixed interior parameter, the smallest constant in a *non-strict* all- N bound is $R_1(a)$ and is attained at $N = 1$.

4 A strictly positive surplus over Lebesgue density

The critical density contains a large cancellation when written as $-\zeta(b)k_a + C_a$. Its residual above one has an unexpectedly clean sign. This sign will be the source of stronger moment inequalities and a sharp lower envelope for Euler errors.

Lemma 4.1 (Critical power balance). *For $0 < a < 1$, $b = 1 - a$, and $T > 0$,*

$$\int_0^T [k_a(T-t) - k_a(T)]t^{b-2} dt = k_a(T) \int_T^\infty t^{b-2} dt = \frac{k_a(T)T^{-a}}{a}. \quad (4.1)$$

Proof. After $t = Tv$, the required scalar integral is

$$\int_0^1 [(1-v)^{a-1} - 1]v^{-a-1} dv = \frac{1}{a}.$$

An antiderivative is $-a^{-1}[(1-v)^a - 1]v^{-a}$. Its lower limit is zero because $[(1-v)^a - 1]v^{-a} = O(v^{1-a})$; its upper limit is $1/a$. Both endpoint singularities of the integrand are integrable. Rescaling proves the identity. $\square$

Set $\phi(t) = t/(e^t - 1)$, which decreases strictly on $(0, \infty)$. Its derivative has numerator $e^t(1-t) - 1 < 0$: that numerator vanishes at zero and has derivative $-te^t < 0$.

Theorem 4.2 (Positive surplus identity). *For every $0 < a < 1$ and $T > 0$, the density in (3.3) satisfies $p_a(T) > 1$. More precisely,*

$$\begin{aligned} p_a(T) - 1 &= \frac{1}{\Gamma(b)} \int_0^T [k_a(T-t) - k_a(T)]t^{b-2}[\phi(t) - \phi(T)] dt \\ &\quad + \frac{k_a(T)}{\Gamma(b)} \int_T^\infty t^{b-2}[\phi(T) - \phi(t)] dt. \end{aligned} \quad (4.2)$$

Both integrals are finite and strictly positive.

Proof. For $0 < b < 1$, the standard subtracted Bose integral is

$$\zeta(b) = \frac{1}{\Gamma(b)} \int_0^\infty t^{b-1} \left(\frac{1}{e^t - 1} - \frac{1}{t} \right) dt. \quad (4.3)$$

For clarity, its continuation can be justified by splitting at one: subtract $1/t$ only on $(0, 1)$ in the usual zeta integral, add $1/(b-1)$, and then use $-\int_1^\infty t^{b-2} dt = 1/(b-1)$. The resulting integral converges at both ends for $0 < b < 1$.

Write $h(t) = 1 + r(t)$, where $r(t) = (e^t - 1)^{-1} - t^{-1}$. The beta convolution of k_a with k_b is $k_1 = 1$. Substituting (4.3) into (3.3) therefore gives

$$p_a(T) - 1 = \frac{1}{\Gamma(b)} \int_0^T [k_a(T-t) - k_a(T)] t^{b-1} r(t) dt \\ - \frac{k_a(T)}{\Gamma(b)} \int_T^\infty t^{b-1} r(t) dt.$$

All terms are now separately integrable. Since $t^{b-1}r(t) = t^{b-2}(\phi(t) - 1)$, the two contributions from the constant -1 cancel by Lemma 4.1. Subtracting the constant $\phi(T)$ against the same balanced signed weight proves (4.2).

On $(0, T)$, $k_a(T-t) > k_a(T)$ because $a < 1$, and $\phi(t) > \phi(T)$. On (T, ∞) the other bracket is also strictly positive. The local behavior at zero is $O(t^{b-1})$, the singularity at T is integrable, and the tail has power t^{b-2} with exponent less than -1 . This proves finiteness and strict positivity. $\square$

An equivalent form, useful for large T , is

$$p_a(T) - 1 = \frac{1}{\Gamma(b)} \int_0^T \frac{[k_a(T-t) - k_a(T)] t^{b-1}}{e^t - 1} dt \\ - \frac{k_a(T)}{\Gamma(b)} \int_T^\infty \frac{t^{b-1}}{e^t - 1} dt. \quad (4.4)$$

The second term is negative, so this version is not by itself a positivity proof. Formula (4.2), not a guessed sign of an asymptotic expansion, supplies that proof.

4.1 Stronger harmonic and Hurwitz moment inequalities

Theorem 4.3 (Strict critical surplus moments). *For $0 < a < 1$, put*

$$Q_n(a) = \frac{1}{a} - \frac{H_{n-1}^{(1-a)}}{n^a} - \frac{1}{n}, \quad n \geq 1. \quad (4.5)$$

Then

$$Q_n(a) = \int_0^1 u^{n-1} [p_a(-\log u) - 1] du. \quad (4.6)$$

Every $\Delta^k Q_n(a)$ is strictly positive, including $k = 0$. Every matrix $(Q_{j+k+1})_{j,k=0}^r$ is positive definite, and $Q_n Q_{n+2} > Q_{n+1}^2$. In particular,

$$\frac{H_{n-1}^{(1-a)}}{n^a} < \frac{1}{a} - \frac{1}{n}, \quad (4.7)$$

$$\frac{H_n^{(1-a)}}{(n+1)^a} - \frac{H_{n-1}^{(1-a)}}{n^a} > \frac{1}{n(n+1)}. \quad (4.8)$$

Proof. Subtract the Lebesgue moment $1/n$ from the density moment in Lemma 3.1. Theorem 4.2 gives a positive, nonzero density on every open subinterval of $(0, 1)$. Hence

$$\Delta^k Q_n = \int_0^1 u^{n-1} (1-u)^k [p_a(-\log u) - 1] du > 0.$$

For any nonzero polynomial P , the integral of $P(u)^2$ against this density is positive. This is precisely positive definiteness of the Hankel matrices. Strict Cauchy–Schwarz with the weighted measure $u^{n-1}[p_a(-\log u) - 1] du$ gives strict log-convexity. The cases $k = 0$ and $k = 1$ give (4.7) and (4.8). The general Hausdorff-moment mechanism is classical; see [8] for the associated generating-function viewpoint. The new input here is the specific positive surplus density. $\square$

The corresponding continuous identity is

$$Q_a(q) = \frac{1}{a} - q^{-a}[\zeta(1-a) - \zeta(1-a, q)] - \frac{1}{q} = \int_0^\infty e^{-qT}[p_a(T) - 1] dT > 0. \quad (4.9)$$

For every $q > 0$ and integer $j \geq 0$, differentiation gives $(-1)^j Q_a^{(j)}(q) > 0$. Exponential domination at infinity and the integrable local density justify all derivatives. This strengthens the prior positivity of the uncorrected critical defect: a full Lebesgue moment may be subtracted while retaining strict complete monotonicity.

There are also two exact normalization identities:

$$\int_0^\infty [p_a(T) - 1] dT = \frac{1}{a}, \quad \int_0^\infty e^{-T}[p_a(T) - 1] dT = \frac{1}{a} - 1. \quad (4.10)$$

For the first, the Hurwitz shift relation and its Taylor expansion at one give $\zeta(b, q) = q^{-b} + \zeta(b) - b\zeta(b+1)q + O(q^2)$ as $q \downarrow 0$. Equation (4.9) becomes $Q_a(q) = a^{-1} - b\zeta(b+1)q^b + O(q^{b+1})$. Monotone convergence against the positive density gives the first identity; the second is $Q_a(1) = 1/a - 1$. In particular $a[p_a(T) - 1] dT$ is a probability measure on $(0, \infty)$ whose exponential moment $\mathbb{E}(e^{-T})$ equals $1 - a$.

4.2 A zero-free difference of polylogarithms

Corollary 4.4 (Surplus Stieltjes factorization). *For $0 < a < 1$ and $z \in \mathbb{C} \setminus [1, \infty)$,*

$$\frac{1-z}{z^2} [F_{a,1-a}(z) - \text{Li}_0(z) + \text{Li}_1(z)] = \int_0^1 \frac{(1-u)[p_a(-\log u) - 1]}{1-zu} du. \quad (4.11)$$

The right side is positive for real $z < 1$ and has strictly positive imaginary part for $\text{Im } z > 0$. Consequently $F_{a,1-a} - \text{Li}_0 + \text{Li}_1$ has no zero on the slit plane except a double zero at zero, whose leading coefficient is $2^{-a} - 1/2$.

Proof. The endpoint function $\text{Li}_0 - \text{Li}_1$ has measure $\delta_1 - du$. Subtract it from (3.4); the resulting signed measure is $(1/a - 1)\delta_1 - [p_a(-\log u) - 1] du$ and has total mass zero. Subtracting $z/(1-z)$ times that zero mass gives (4.11). The numerator defines a strictly positive measure. The imaginary part of its resolvent is

$$(\text{Im } z) \int_0^1 \frac{u(1-u)[p_a(-\log u) - 1]}{|1-zu|^2} du,$$

which is strict. The real case is immediate from positive denominators. Finally, comparing the z^2 coefficients yields $2^{-a} - 1/2 > 0$. $\square$

This is a zero-free theorem for a *difference* of functions; it is not a reannouncement of the earlier zero-free theorem for $F_{a,b}$. Nor does it decide angular uniqueness below $a + b = 1$.

5 Finite Euler depth: endpoints, lower bounds, and a reversal

Proposition 5.1 (Endpoint error identities and sharp lower envelope). *For each fixed $N \geq 1$, R_N extends continuously to $[0, 1]$, with*

$$R_N(0) = J_N^* := \int_0^1 \frac{q_N(u)}{1-u} du, \quad R_N(1) = I_N := \int_0^1 q_N(u) du. \quad (5.1)$$

For every $0 < a < 1$, $R_N(a) > I_N$. Thus I_N is the sharp lower envelope over the critical line.

Proof. For fixed $T > 0$, (3.3) and the endpoint beta laws give

$$\lim_{a \downarrow 0} p_a(T) = \frac{1}{1 - e^{-T}}, \quad \lim_{a \uparrow 1} p_a(T) = 1.$$

Indeed, $-\zeta(1-a)/\Gamma(a) \rightarrow 1$ at zero and tends to $1/2$ at one; the respective beta averages tend to $h(T)$ and $1/2$. The coefficient $-\zeta(1-a)/\Gamma(a)$ is bounded on the closed parameter interval after these continuous extensions. Consequently, with an absolute constant,

$$0 < p_a(T) \leq C(1 + T^{-1}) \quad (T > 0, 0 < a < 1). \quad (5.2)$$

For $T = -\log u$, the right side is at most $C'(1-u)^{-1}$. The weight $q_N(u)/(1-u)$ is integrable for $N \geq 1$, so dominated convergence proves the endpoint limits and continuity. Strict surplus positivity proves $R_N(a) > I_N$. $\square$

The limiting density at $a = 0$ has infinite total mass. Only its filtered integral in (5.1) is finite; it must not be substituted as a finite signed measure into (3.4).

There are elementary finite recurrences for these endpoint values. Let

$$B_j^{\text{pol}} = \int_0^1 (1-u^2)^j du = \frac{4^j (j!)^2}{(2j+1)!}. \quad (5.3)$$

Then

$$I_0 = \frac{\pi}{4}, \quad I_{N+1} = 2I_N - B_N^{\text{pol}} \quad (N \geq 0), \quad (5.4)$$

$$J_1^* = \frac{\pi}{4} + \frac{\log 2}{2}, \quad J_{N+1}^* = 2J_N^* - B_{N-1}^{\text{pol}} - \frac{1}{2N} \quad (N \geq 1). \quad (5.5)$$

To prove these, use $q_{N+1} = 2q_N - (1-u^2)^N$ and $(1-u^2)^N/(1-u) = (1+u)(1-u^2)^{N-1}$. The recurrences are exact identities, but their subtractive form may be poor for floating-point evaluation at large N .

Theorem 5.2 (A certified finite-depth reversal). *Strictly,*

$$0.084208864068217362453965253610 < R_{128}(1/4) < 0.084208864068217362453965253612, \quad (5.6)$$

$$0.084523512538326867477886761599 < R_{128}(1/2) < 0.084523512538326867477886761601. \quad (5.7)$$

In particular $R_{128}(1/2) > R_{128}(1/4)$, whereas $R_1(1/2) < R_1(1/4)$.

Proof. The last inequality is also an instance of Theorem 3.2. The two displayed enclosures follow from the standard-library-only certificate `code/certify.py`. Its finite arithmetic and tail contract are described in Section 8; a verifier replays integer-root acceptance inequalities and outward interval operations. In particular the assertion is not inferred from decimal agreement between two floating-point calculations. $\square$

This counterexample refutes a *stronger extrapolation*, namely that $R_N(a)$ decreases in a for each N . The incoming report does not assert that extrapolation. Its actual conjecture concerns R_1 , or equivalently the optimal constant after taking the supremum over N ; that conjecture is proved, not contradicted, here.

6 Large-logarithm structure of the critical density

Put $b = 1 - a$ and define the positive coefficients

$$\alpha_j(b) = \frac{(b)_j^2 \zeta(b+j)}{j! \Gamma(1-b)}, \quad j \geq 1, \quad A(b) = \alpha_1(b) = \frac{b^2 \zeta(1+b)}{\Gamma(1-b)}. \quad (6.1)$$

Here $(b)_j = b(b+1) \cdots (b+j-1)$ is a rising factorial.

Theorem 6.1 (All-orders density expansion). *For fixed $0 < b < 1$ and every integer $M \geq 1$,*

$$p_{1-b}(T) = 1 + \sum_{j=1}^M \alpha_j(b) T^{-b-j} + O(T^{-b-M-1}) \quad (T \rightarrow \infty). \quad (6.2)$$

The remainder is uniform when b ranges over a compact subinterval of $(0, 1)$. This is a Poincaré asymptotic expansion, not a claim of convergence for a fixed T or a globally signed truncation theorem.

Proof. Write (4.4) as

$$p_{1-b}(T) - 1 = \frac{T^{-b}}{\Gamma(1-b)\Gamma(b)} \left\{ \int_0^T \frac{[(1-t/T)^{-b} - 1]t^{b-1}}{e^t - 1} dt - \int_T^\infty \frac{t^{b-1}}{e^t - 1} dt \right\}. \quad (6.3)$$

On $0 \leq t \leq T/2$, expand $(1-t/T)^{-b} - 1 = \sum_{j=1}^M (b)_j (t/T)^j / j! + O((t/T)^{M+1})$. The remainder constant is uniform on compact b -sets. Integrating termwise and extending each exponentially decaying moment to infinity introduces exponentially small errors. The moments are $\Gamma(b+j)\zeta(b+j)$, so their coefficients are exactly (6.1). On $T/2 < t < T$, the integrable singularity $(T-t)^{-b}$ is multiplied by an exponentially small Bose factor; its contribution and the last integral are smaller than every displayed power, uniformly on the same compact sets. This proves the result. $\square$

The maximizer will approach a parameter endpoint. A merely compact-uniform statement is insufficient for optimizing over all a . The following estimate supplies the missing endpoint uniformity.

Lemma 6.2 (A uniform first-term estimate). *There are absolute constants C, T_0 such that for all $0 < b < 1$ and $T \geq T_0$,*

$$\left| p_{1-b}(T) - 1 - A(b)T^{-1-b} \right| \leq CA(b)T^{-2-b} + Ce^{-T/2}. \quad (6.4)$$

One may take $T_0 = 2$ with a sufficiently large absolute C .

Proof. For $0 \leq s \leq 1/2$ and $0 < b < 1$, Taylor's formula gives

$$0 \leq (1-s)^{-b} - 1 - bs \leq 8bs^2.$$

Use this in (6.3) on $(0, T/2)$. Its leading integral, when extended to infinity, is $A(b)T^{-1-b}$. The remainder is at most

$$\frac{8b\Gamma(b+2)\zeta(b+2)}{\Gamma(1-b)\Gamma(b)}T^{-b-2} = 8(b+1)\frac{\zeta(b+2)}{\zeta(b+1)}A(b)T^{-b-2} \leq 16A(b)T^{-b-2}.$$

The omitted tail of the leading moment is $O(e^{-T/2})$ uniformly: for $x \geq 1$ and $0 < b < 1$, $\int_x^\infty t^b e^{-t} dt \leq 2x^b e^{-x}$, while reciprocal gamma factors are bounded on the parameter interval.

For the singular contribution on $T/2 < t < T$, use $(e^t - 1)^{-1} \leq (1 - e^{-1})^{-1}e^{-t}$ and retain the complete beta weight rather than bounding its endpoint exponent separately. The term containing $(1 - t/T)^{-b}$ is bounded by a constant times

$$\frac{e^{-T/2}}{\Gamma(1-b)\Gamma(b)} \int_0^T (T-t)^{-b} t^{b-1} dt = e^{-T/2}.$$

The remaining nonsingular tails in (6.3) are bounded by the same type of estimate, using $T \geq 2$. This avoids a spurious factor diverging as $b \uparrow 1$. Combining the bounds proves (6.4). $\square$

7 Uniform Euler asymptotics and the drifting extremizer

For the remainder of this section set

$$L = \frac{1}{2} \log N, \quad \ell = \log L, \quad B_N = \frac{\sqrt{\pi}}{2\sqrt{N}}. \quad (7.1)$$

Statements using ℓ concern $N \rightarrow \infty$; they are not proposed as accurate low-depth formulas.

7.1 A uniform localization lemma

Lemma 7.1 (Gaussian logarithmic averaging). *Uniformly for $1 \leq s \leq 3$,*

$$\int_0^{N^{-1/4}} q_N(u)(-\log u)^{-s} du = B_N L^{-s} (1 + O(L^{-1})). \quad (7.2)$$

Also

$$I_N = B_N \left(1 - \frac{7}{8N} + O(N^{-2}) \right). \quad (7.3)$$

Proof. Set $u = t/\sqrt{N}$. Then $-\log u = L - \log t$ and $0 < t < N^{1/4}$. On this interval

$$q_N(t/\sqrt{N}) = e^{-t^2} + O\left(N^{-1}(t^2 + t^4)e^{-t^2}\right),$$

in the sense of an absolute pointwise bound with a fixed constant for large N . This follows from $-v - v^2/[2(1-v)] \leq \log(1-v) \leq -v$ and $1 - e^{-x} \leq x$, with $v = t^2/N \leq N^{-1/2}$. On $e^{-L/2} \leq t \leq e^{L/2}$, the mean-value theorem gives, uniformly for $1 \leq s \leq 3$,

$$\left| \left(1 - \frac{\log t}{L} \right)^{-s} - 1 \right| \leq C \frac{|\log t|}{L}.$$

The Gaussian logarithmic moment is integrable. On $0 < t < e^{-L/2}$ the normalized factor lies between zero and one and the interval has length $e^{-L/2}$. Gaussian tails above $e^{L/2}$ are negligible. This proves (7.2) with constants independent of s .

For (7.3), expand one term further on a central interval and bound its complement by Gaussian decay:

$$q_N(t/\sqrt{N}) = e^{-t^2} \left[1 - \frac{t^2 + t^4/2}{N} + O(N^{-2}P(t)) \right],$$

where a fixed polynomial P suffices on a growing central interval. The normalized half-Gaussian moments of t^2 and t^4 are $1/2$ and $3/4$. Hence the coefficient is $1/2 + (3/4)/2 = 7/8$. Taylor remainders on, for example, $t \leq N^{1/12}$ and exponential bounds on the complement justify the integrated $O(N^{-2})$ term. $\square$

Theorem 7.2 (Uniform critical Euler expansion). *Uniformly for every $0 < b < 1$ as $N \rightarrow \infty$,*

$$R_N(1-b) = I_N + B_N A(b) L^{-1-b} (1 + O(L^{-1})) + O(N^{-3/4}). \quad (7.4)$$

More explicitly, the absolute difference after subtracting $I_N + B_N A(b) L^{-1-b}$ is at most $C B_N A(b) L^{-2-b} + C N^{-3/4}$ with an absolute constant. In particular,

$$\sup_{0 < a < 1} \left| \frac{R_N(a)}{B_N} - 1 \right| \rightarrow 0. \quad (7.5)$$

Proof. By (3.9), the difference $R_N - I_N$ is the integral of $q_N(u)[p_a(-\log u) - 1]$. First restrict to $u \leq N^{-1/4}$, so that $T = -\log u \geq L/2 \geq 2$ for large N . Apply Lemma 6.2. The leading term is evaluated by Lemma 7.1 with $s = 1 + b$; the algebraic error uses $s = 2 + b$. The exponential density error integrates to at most

$$C \int_0^1 q_N(u) u^{1/2} du \leq C \int_0^\infty e^{-Nu^2} u^{1/2} du = O(N^{-3/4}).$$

For the complementary interval, use (5.2) and

$$\frac{q_N(u)}{1-u} = \frac{1+u}{1+u^2} (1-u^2)^{N-1} \leq 2e^{-(N-1)N^{-1/2}} \quad (u \geq N^{-1/4}).$$

Its total contribution is exponentially small uniformly in a . This proves the explicit remainder. Finally, $A(b)$ is bounded on $(0, 1)$ and extends continuously with value zero at both endpoints, so (7.5) follows from (7.3). $\square$

7.2 Fixed-parameter coefficients to all orders

Let

$$m_r = \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-t^2} (\log t)^r dt = \frac{\Gamma^{(r)}(1/2)}{2^r \Gamma(1/2)}. \quad (7.6)$$

In particular $m_0 = 1$, $m_1 = -\gamma/2 - \log 2$, and $m_2 = m_1^2 + \pi^2/8$. For $m \geq 1$ define the explicit zeta-gamma coefficient

$$D_m(b) = \sum_{j=1}^m \alpha_j(b) \frac{(b+j)_{m-j}}{(m-j)!} m_{m-j}. \quad (7.7)$$

Corollary 7.3 (All logarithmic orders). *For fixed $0 < b < 1$ and any integer $M \geq 1$,*

$$R_N(1-b) = I_N + B_N \sum_{m=1}^M D_m(b) L^{-b-m} + O(B_N L^{-b-M-1}). \quad (7.8)$$

The result is uniform on compact b -subintervals of $(0, 1)$. The first three coefficients are

$$\begin{aligned} D_1 &= \alpha_1, \\ D_2 &= \alpha_2 + (b+1)m_1\alpha_1, \\ D_3 &= \alpha_3 + (b+2)m_1\alpha_2 + \frac{1}{2}(b+1)(b+2)m_2\alpha_1. \end{aligned}$$

Proof. Insert Theorem 6.1 into the localized error integral. For each density term expand

$$(L - \log t)^{-b-j} = L^{-b-j} \sum_{r=0}^{M-j} \frac{(b+j)_r}{r!} \frac{(\log t)^r}{L^r} + \text{remainder}.$$

On $|\log t| \leq L/2$ Taylor's remainder is controlled by the next Gaussian logarithmic moment. The omitted small- t interval has exponentially small length, and the large- t interval has Gaussian decay. The q_N replacement error is polynomially small in N , hence smaller than every fixed inverse logarithmic order. Integrate termwise using (7.6), and collect terms with $j+r=m$ to obtain (7.7). Compact-uniform constants follow from the same estimates. $\square$

This finite coefficient formula gives an entire family of proved asymptotic identities. The positivity of the density coefficients α_j does not imply that every D_m is positive: logarithmic Gaussian moments can have either sign.

7.3 Maximization over the critical line

Let $R_N(0)$ and $R_N(1)$ have their continuous endpoint meanings from Proposition 5.1, and define

$$C_N = \max_{0 \leq a \leq 1} R_N(a). \quad (7.9)$$

A maximizer exists by continuity. Its value is also the supremum over the open critical line.

Theorem 7.4 (Drift of every worst-case parameter). *For all sufficiently large N , every maximizer a_N in (7.9) lies in $(0, 1)$. As $N \rightarrow \infty$,*

$$(1 - a_N)\ell \rightarrow 1, \quad (7.10)$$

$$C_N = B_N \left[1 + \frac{1 + o(1)}{eL\ell} \right]. \quad (7.11)$$

No uniqueness of the finite- N maximizer is assumed or concluded.

Proof. Extend A to $[0, 1]$ by zero at both endpoints. Since the zeta pole has residue one and $\Gamma(1) = 1$,

$$A(b) = b(1 + O(b)) \quad (b \downarrow 0), \quad A(b) \leq Cb \quad (0 \leq b \leq 1). \quad (7.12)$$

Define $f_\ell(b) = A(b)e^{-\ell b}$. We claim

$$\max_{0 \leq b \leq 1} f_\ell(b) \sim \frac{1}{e\ell}, \quad \ell b_\ell \rightarrow 1 \quad (7.13)$$

for every maximizing b_ℓ and also every asymptotically maximizing sequence. Testing $b = 1/\ell$ gives the lower estimate. For $t = \ell b$ in a fixed compact interval, $\ell f_\ell(t/\ell) \rightarrow te^{-t}$ uniformly. The upper bound in (7.12) makes the contribution for large t uniformly small, and the same bound makes

the contribution for small t small. The function te^{-t} has a unique maximum $1/e$ at $t = 1$, proving (7.13).

Theorem 7.2, after multiplication by L/B_N , is

$$\frac{L}{B_N}(R_N(1-b) - I_N) = f_\ell(b) + O(f_\ell(b)/L) + O(LN^{-1/4}), \quad (7.14)$$

uniformly on the open interval. At $a = 1$ the left side is zero. At $a = 0$, a direct substitution $u = t/\sqrt{N}$ gives

$$J_N^* - I_N = \int_0^1 q_N(u) \frac{u}{1-u} du = \frac{1}{2N} + O(N^{-3/2}). \quad (7.15)$$

Thus (7.14) is valid at the endpoints as well with an error of the stated or smaller size. Its errors are $o(1/\ell)$, whereas the largest surrogate value is asymptotic to $1/(e\ell)$. Every true maximizer is therefore asymptotically maximizing for the surrogate and obeys (7.10). It cannot be an endpoint for large N . The maximum excess is $B_N/(eL\ell)(1 + o(1))$. Finally $I_N/B_N = 1 + O(N^{-1})$ proves (7.11). $\square$

A useful consequence of the same proof is that, for *every fixed* $0 < a < 1$, $R_N(a) > R_N(0)$ once N is sufficiently large (depending on a). Indeed its positive excess over I_N has order $N^{-1/2}L^{-1-b}$, while the $a = 0$ excess has order N^{-1} . This supplies an analytic explanation for the finite-depth reversal, independent of the particular certified pair.

7.4 A Stieltjes-constant correction

Use the convention

$$\zeta(1+b) = \frac{1}{b} + \gamma - \gamma_1 b + O(b^2)$$

for the first Stieltjes constant γ_1 [6]. The gamma expansion [7] gives

$$A(b) = b - db^3 + O(b^4), \quad d = \gamma_1 + \frac{\gamma^2 + \zeta(2)}{2} = 0.916240149844295830534809275626 \dots \quad (7.16)$$

In particular the quadratic coefficient vanishes exactly.

Theorem 7.5 (Refined drift and optimal excess). *For every maximizer a_N and the notation (7.1),*

$$1 - a_N = \frac{1}{\ell} - \frac{2d}{\ell^3} + O(\ell^{-4}), \quad (7.17)$$

$$C_N = B_N \left[1 + \frac{1}{eL\ell} \left\{ 1 - \frac{d}{\ell^2} + O(\ell^{-3}) \right\} \right]. \quad (7.18)$$

The location estimate applies to all maximizing choices, even if some finite depth has more than one maximizer.

Proof. Expansion of the reciprocal gamma factor in (6.1) gives

$$\Gamma(1-b)^{-1} = 1 - \gamma b + \frac{1}{2}(\gamma^2 - \zeta(2))b^2 + O(b^3),$$

which proves (7.16). Near $b = 0$ it follows that

$$\frac{A'(b)}{A(b)} = \frac{1}{b} - 2db + O(b^2).$$

By (7.13), all surrogate maximizers eventually lie in $3/(4\ell) < b < 5/(4\ell)$. On that interval $f''_\ell(b)$ is negative and bounded above by $-c\ell$ for some $c > 0$: differentiating $A(b)e^{-\ell b}$ and using $A(b) = b + O(b^3)$ gives $f''_\ell(b) = e^{-\ell b}(\ell^2 b - 2\ell + O(\ell^{-1}))$. Hence there is one surrogate maximum b_ℓ^* in this interval. Its equation $A'/A = \ell$ implies

$$b_\ell^* = \ell^{-1} - 2d\ell^{-3} + O(\ell^{-4}), \quad f_\ell(b_\ell^*) = \frac{1}{e\ell}(1 - d\ell^{-2} + O(\ell^{-3})).$$

It remains to transfer location, not just value, to the true error. Theorem 7.4 places every true maximizer in the same interval for large N . On it the uniform error in (7.14) is $O((\ell L)^{-1} + LN^{-1/4})$. Comparing the true maximum with the value at b_ℓ^* bounds the surrogate value loss by twice this error. Strict concavity therefore implies

$$|(1 - a_N) - b_\ell^*|^2 \leq C \left(\frac{1}{\ell^2 L} + \frac{LN^{-1/4}}{\ell} \right).$$

Since $L = e^\ell$ and $N = e^{2L}$, this location difference is smaller than every fixed inverse power of ℓ . This proves (7.17) without differentiating the true error remainder. The same uniform comparison transfers the surrogate maximum value with an error smaller than any displayed inverse- ℓ order; (7.3) then gives (7.18). $\square$

Remark 7.6 (Why the two extremal statements are compatible). The sharp constant after optimizing over *both* a and N is C_* , approached at $a \downarrow 0, N = 1$. At a large *prescribed* depth N , the maximum instead occurs at an interior parameter approaching $a = 1$. The rate is inverse $\log \log N$, more precisely (7.17) with $\ell = \log(\frac{1}{2} \log N)$. This slow rate makes the limiting formulas poor low-depth predictors. The statements concern different optimizations; neither reverses or weakens Theorem 3.2.

8 Exact certificates and independent computations

8.1 Arithmetic contract of the rational verifier

All certified quantities use the already proved bound $0 < E_K - g < (1/a)2^{-K}$ with $K = 256$. In particular, the verifier does not assume the new constant C_* to certify its examples. The finite sum is formed with the identity

$$E_N = 2^{-N} \sum_{n=0}^{N-1} (-1)^n f(n) \sum_{j=n+1}^N \binom{N}{j}. \quad (8.1)$$

To prove it, substitute the finite definition of d_k and use

$$\sum_{k=n}^{N-1} 2^{N-k-1} \binom{k}{n} = \sum_{j=n+1}^N \binom{N}{j}.$$

Pascal's identity proves the coefficient relation by induction. This avoids constructing a large interval difference table.

For an inverse rational power $n^{-p/q}$, choose $S = 10^{210}$ and compute an integer proposal r for the q th root of $n^p S^q$. The verifier accepts it only after checking

$$r^q \leq n^p S^q < (r+1)^q. \quad (8.2)$$

If the left equality does not hold, $S/(r+1) < n^{-p/q} < S/r$. In either case these rational endpoints are rounded outward to the grid $S^{-1}\mathbb{Z}$ using integer division. Positive products, signed linear combinations, and division by positive integers are also rounded outward. The root-finding iteration is a proposal method; (8.2), not the iteration history, is its guarantee.

An enclosure for g is obtained by subtracting $(1/a)2^{-256}$ from the lower endpoint of the finite E_{256} enclosure and leaving the upper endpoint unchanged. Subtract this g interval from the finite E_N interval and multiply by 2^N to obtain R_N . At $N = 128$ the analytic tail width is at most $(1/a)2^{-128}$, well below the displayed 30-place comparison. Finally the program tests the human-readable rational endpoints in (5.6)–(5.7) directly by integer cross multiplication.

8.2 Recorded exact checks

The supplied run passes 2,555 integer-root acceptance checks. It certifies Gaussian values and six scaled-error depths for each of $a = 1/10, 1/4, 1/2, 3/4, 9/10$, hence 30 positive-error enclosures. It also checks 56 strict surplus finite differences for each parameter, for 280 such tests in total. These finite tests illustrate Theorem 4.3; the theorem’s all-index content comes from the positive integral proof, not from extrapolating the test grid. The exact difference certificate is

$$\begin{aligned} 0.000314648470109505023921507988 &< R_{128}(1/2) - R_{128}(1/4) \\ &< 0.000314648470109505023921507989. \end{aligned} \tag{8.3}$$

The JSON stores outward intervals rather than unsupported point values. A separate SymPy script checks eight algebraic identities and 528 finite instances of the binomial weight identity. It does not check analytic continuation, dominated convergence, or the uniformity of asymptotic remainders; those arguments are in the article.

8.3 Independent numerical diagnostics

The floating-point program uses 180 decimal digits for finite Euler calculations. Five Gaussian values are separately checked by beta–gamma quadrature, which evaluates (2.3) rather than an Euler difference table. Nine density tests compare the beta average (3.3) with the regularized Bose expression (4.4) at $a = 1/4, 1/2, 3/4$ and $T = 2, 10, 50$. These checks also compare one- and two-term density asymptotics. They are diagnostics, not rigorous quadrature error bounds.

A grid scan followed by bounded local optimization gives the following *proposals*; uniqueness and global optimality at these particular depths are not certified by the optimizer.

N	proposed maximizing a	proposed max R_N
8	0.10599483	0.359993461628
16	0.21532658	0.249985265562
32	0.29829873	0.173968759177
64	0.36226668	0.121253639850
128	0.41230987	0.084666552368
256	0.45206073	0.059234316775

The movement is consistent with the proved drift, but the proof is independent of the table. The program additionally optimizes the surrogate $A(b)e^{-\ell b}$ for larger ℓ ; those surrogate results are explicitly labelled and are not presented as astronomical- N computations of the original error.

8.4 Reproduction

From the package directory, run

```
python code/certify.py
python code/symbolic_checks.py
python code/diagnostics.py
python code/build_article.py
```

Only the first command is needed to replay the rational certificates; it uses the Python standard library. The optional diagnostics require mpmath, NumPy, SciPy, and SymPy at the versions listed in `requirements-diagnostics.txt`. The build requires a local `pdflatex`. All scripts resolve their paths relative to their own location, accept no external numerical oracle, and write their results to the package's `data/` directory. Run them on a copy when preserving a delivered evidence snapshot.

9 Audit conclusions and integration

9.1 What should change in the claim ledger

The incoming report's Conjecture 12.3 can be promoted to a theorem with Theorem 3.2 as its proof reference. Its question about sharp N -dependence is answered by Theorems 7.2, 7.4, and 7.5. The new proof route relies on the existing positive double kernel and critical measure, not on any unproved numerical relation among special values.

No error in the *stated* critical measure or Euler bound of the reviewed incoming report was established. The relevant corrections are to possible extensions or editorial wording: do not replace the critical measure by a density without its atom; do not describe its boundary series as ordinarily convergent; do not omit the factor X from the new rate-one beta–gamma coupling; and do not extend the proved decrease of R_1 to the functions R_N individually. The last overextension is now excluded by an exact certificate.

The asymptotic expansion (6.2) must not be advertised as a convergent global series or as a signed remainder theorem. For parameter optimization use (6.4), which is uniform up to both parameter endpoints, rather than the compact-uniform fixed-order expansion. Finally, the stationary point of the surrogate is not the exact finite- N optimizer; Theorem 7.5 proves their asymptotic proximity and the numerical records keep them separate.

9.2 Suggested repository placement

The complete delivery can be placed under

Analysis/Polylogarithms/docs/reports/critical-euler-transport/.

Its thematic predecessor is the incoming real-order-threshold report. The manuscript insertion supplied in `integration/` uses the prefix `criticaleuler:` and is a summary with references to the full proofs, not a replacement for them. The main proof of the sharp constant belongs near the signed-kernel and certified-evaluation material; the positive surplus moments connect naturally to the Hurwitz-jet chapters. Intake should preserve the original delivery and record the change of conjecture status in the programme ledger.

The existing S_6 identity, subcritical angular uniqueness, and the broad normalized-radius conjecture are not resolved by this article. No new independence, minimal-depth, or transcendence assertion follows from these positive kernels. This distinction is especially important because a new numerical evaluation method does not by itself reduce a constant to a prescribed arithmetic period basis.

10 Further research questions

10.1 Uniqueness and monotonic motion of the finite-depth maximizer

The asymptotic theory localizes every maximizer but does not exclude several extremely close maxima at a finite depth. The numerical samples suggest a stronger statement.

Conjecture 10.1 (Eventual unimodality). *For every sufficiently large integer N , $a \mapsto R_N(a)$ has exactly one critical point in $(0, 1)$, a strict global maximum. The maximizing outer order increases strictly with N once N is sufficiently large.*

A route to proof is to differentiate the balanced density identity in the shape parameter and establish uniform derivative remainders. The function-value argument in Theorem 7.5 deliberately avoids such estimates and therefore cannot supply uniqueness. A useful finite problem is to determine the first depth at which the maximum leaves $a = 0$, with certified endpoint derivative signs and an exhaustive zero count for the derivative.

10.2 An explicit nonasymptotic optimal-depth bound

Theorem 7.2 gives absolute uniform constants but does not optimize or enumerate them. An implementable bound of the form

$$R_N(a) \leq \frac{\sqrt{\pi}}{2\sqrt{N}} \left(1 + \frac{K}{\log N \log \log N} \right)$$

with explicit K and a verified threshold would improve practical certificates for all critical orders at once. Below that threshold one could combine the sharp all-depth constant with a finite exact parameter enclosure. This requires interval control over a continuum of a , not merely a table of rational sample orders.

10.3 The exact radial domain of order transport

Determine for which pairs (w, ρ) the strict decrease of $G_{a, w-a}(\rho)$ holds for all $a \in (0, w)$. Our theorem covers $0 < \rho \leq \sqrt{3}$ for every $w > 0$. Beyond that radius the pointwise kernel changes derivative sign, but the beta–gamma average may preserve the comparison in a larger domain. A proof or a rigorously enclosed counterexample should distinguish these two mechanisms.

10.4 Other Gaussian filters and cyclotomic points

The filter $(1 - u^2)^N / (1 + u^2)$ is particularly suited to the Gaussian point. Root-of-unity filters of higher order lead to different localized moments and may change the worst-order drift. It is natural to ask whether balanced critical densities give similarly sharp constants and explicit Stieltjes-constant corrections for Eisenstein or other cyclotomic evaluations. The positivity of the present filter cannot simply be presumed for a complex-valued generalization.

10.5 Higher-depth critical defects

Replacing $H_{n-1}^{(b)}$ by nested harmonic sums should introduce more than one critical balance. A precise objective is to identify the largest elementary positive moment measure that can be subtracted from the corresponding defect while retaining positivity. At depth two the answer supplied here includes the entire Lebesgue measure. At higher depth, an endpoint atom and an unproved sign pattern must not be confused with a fully established signed Hausdorff representation.

10.6 Remaining arithmetic and geometric questions

The S_6 reduction and the subcritical angular-zero conjecture remain separate worthwhile targets. The present surplus factorization may help compare critical functions, but it does not construct a finite signed measure below the known threshold. On the computational side, the finite integer-root verifier is a small candidate for formal verification; the analytic layer would additionally require beta-gamma integration, signed measure transforms, and uniform asymptotics.

Conclusion

The critical constant is now determined by a strict beta-order comparison, not by numerical optimization. A balanced Bose-kernel identity reveals a positive density surplus, giving stronger harmonic moment inequalities and a new zero-free difference factorization. That same surplus explains why the worst finite-depth Euler parameter moves in the opposite direction from the endpoint controlling the sharp all-depth constant. The uniform error theorem and its Stieltjes-constant refinement identify this movement quantitatively, while a rational certificate exhibits the reversal at a concrete finite depth. These results are a continuation of the inspected real-order programme, with unresolved arithmetic and geometric questions left explicitly separate.

A Claim-status summary

Claim	Evidence and status
Double kernel; critical measure; Euler remainder	Prior analytic identities, rederived where needed and attributed.
Fixed-total-order transport and comparisons	Proved in Section 2; imaginary radii through $\sqrt{3}$ only.
Incoming Conjecture 12.3	Proved, including strict monotonicity and sharp uniform constant.
Strict surplus density and positive moments	Proved by a balanced integral, with all-index and continuous-parameter consequences.
Zero-free surplus difference	Proved for the principal branch on the slit plane.
Uniform Euler expansion and extremizer drift	Proved with endpoint-uniform error; no uniqueness premise.
Refined Stieltjes-constant coefficient	Proved by surrogate comparison and curvature, not a fitted numerical coefficient.
Finite-depth reversal	Exact outward rational certificate using a proved infinite-tail bound.
Quadrature and numerical optimizer tables	Diagnostics only; no interval-certified global optimum claim.

Claim	Evidence and status
Eventual unimodality and monotone maximizing parameter	Conjectural.
S_6 , subcritical angular uniqueness, broad normalized-radius claim	Not settled here.
Global novelty; proof-assistant formalization; exhaustive source audit	Not claimed.

References

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